

1. Administrative & Study Identification

Official Title: A Double-blind, Randomized, Placebo- and Active-Comparator Controlled Study to Evaluate the Efficacy of Inclisiran as Monotherapy in Patients With Primary Hypercholesterolemia Not Receiving Lipid-Lowering Therapy **Study Title:** Efficacy and Safety of Inclisiran as Monotherapy in Patients With Primary Hypercholesterolemia Not Receiving Lipid-lowering Therapy. **Short Title/Acronym:** VictORION-Mono (V-Mono) **Protocol Number:** CKJX839D12304 **NCT Number:** NCT05763875 **Posting ID:** 576102687985321766 **Trial Status:** Completed **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Inclisiran (Leqvio), small interfering RNA (siRNA) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To determine if the study treatment, inclisiran, in comparison to placebo and ezetimibe, effectively reduces Low-Density Lipoprotein Cholesterol (LDL-C) as measured by percentage change from baseline to Day 150. * **Secondary Objectives:** To evaluate the absolute change in LDL-C, the percentage change in proprotein convertase subtilisin/kexin type 9 (PCSK9) from baseline to Day 150, the percentage change in Non-High-Density Lipoprotein Cholesterol (Non-HDL-C), and to evaluate overall safety and tolerability.

2.2 Background & Rationale To address the need for effective LDL-C lowering in patients with primary hypercholesterolemia who are not currently receiving lipid-lowering therapy (LLT) and have a 10-year Atherosclerotic Cardiovascular Disease (ASCVD) risk of less than 7.5%. Inclisiran is a first-in-class siRNA therapy designed to reduce the production of PCSK9 in the liver, thereby increasing the liver's ability to clear LDL-C from the blood, potentially improving medication adherence via an infrequent dosing schedule.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled and Active-Comparator (Ezetimibe) * **Blinding:** Double-blind * **Randomization:** 2:1:1 ratio (Inclisiran : Ezetimibe : Placebo) * **Trial Status:** Completed

3.2 Planned Enrollment & Duration * **Target N**: 350 adult participants * **Number of Sites**: Multicenter (Global) * **Estimated Study Start**: February 2023 * **Estimated Primary Completion**: June 6, 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Informed consent signed prior to participation in the study. 2. Fasting LDL-C of ≥ 100 mg/dL but < 190 mg/dL. 3. Fasting triglycerides ≤ 400 mg/dL. 4. 10-year ASCVD risk score $< 7.5\%$ and not on any lipid-lowering therapy within 90 days of screening.

4.2 Exclusion Criteria 1. History of ASCVD. 2. Diabetes mellitus or fasting plasma glucose of ≥ 7.0 mmol/L or HbA1c $\geq 6.5\%$. 3. Secondary hypercholesterolemia, e.g., hypothyroidism (TSH above the upper limit of normal).

5. Intervention & Treatment Plan

5.1 Investigational Regimen & Comparators * **Investigational Drug**: Inclisiran sodium 300 mg, administered as an initial dose on Day 1, followed by a second dose at 3 months (Day 90). * **Active Comparator**: Ezetimibe (ATC Code: C10AX09; Trade Names: Vytorin, Zetia, Lypqozet, Roszet, Nexlizet, Liptruzet). * **Route of Administration**: Subcutaneous injection for Inclisiran. * **Duration of Treatment**: 150 $\pm$ 5 days for the double-blind treatment period.

5.2 Concomitant & Prohibited Therapy * **Permitted**: Standard medications for comorbidities not affecting lipid metabolism. * **Prohibited**: Concomitant use of any other lipid-lowering therapy (e.g., statins, PCSK9 monoclonal antibodies, or non-study ezetimibe).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, ASCVD risk assessment, Fasting Lipid Labs
Baseline	Day 1	Randomization (2:1:1), First Dose Administration	Interim
Day 90	Second Dose Administration (Inclisiran, Ezetimibe, or Placebo)	End of Study	Day 150 $\pm$ 5 Primary Endpoint Evaluation (Fasting LDL-C and PCSK9 labs)
Safety Follow-Up	Day 180	Safety Follow-up / End of Study visit	30+5 days after the Day 150 visit

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Percentage change in Low-density Lipoprotein Cholesterol (LDL-C) from baseline to Day 150 (Inclisiran arm versus Ezetimibe and placebo). * **Measurement Method:** Central Laboratory Analysis of fasting lipid panels. * **Estimands of Interest:** * **Monotherapy Estimand:** Inclisiran as monotherapy compared to the use of comparator. This estimand uses a hypothetical strategy where participants who permanently discontinued treatment, died, or used other LLTs were handled in a hypothetical scenario of what would have happened if the intercurrent event did not happen. * **Treatment-policy Estimand:** Inclisiran as monotherapy compared to the use of comparator with or without other lipid-lowering therapies (LLTs) added. This estimand ignored the use of other LLTs and treatment discontinuation. Deaths were handled as an unfavorable outcome using a composite variable strategy.

7.2 Secondary & Exploratory Endpoints * Absolute change in LDL-C from baseline to Day 150. * Percentage change in proprotein convertase subtilisin/kexin type 9 (PCSK9) levels from baseline to Day 150. * Percentage change in Non-High-Density Lipoprotein Cholesterol (Non-HDL-C) from baseline to Day 150.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous collection from informed consent through the safety follow-up. Monitored for events such as injection site reactions, arthralgia, and bronchitis.
 - **Serious Adverse Events (SAEs):** Standard protocol timeline reporting (typically within 24 hours of investigator awareness).
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to review ongoing safety signals.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary efficacy (Inclisiran vs. Placebo and vs. Ezetimibe); Safety Set for all participants receiving ≥ 1 dose of study drug.
 - **Sample Size Justification:** $N = 350$ (randomized 2:1:1) to provide statistical power to detect a clinically meaningful superior reduction in LDL-C at Day 150 for inclisiran compared to placebo and ezetimibe ($\alpha = 0.05$).
 - **Handling of Missing Data:** Mixed-effect Model Repeated Measure (MMRM) and/or Multiple Imputation methods utilized for missing LDL-C values at Day 150.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine remote and onsite monitoring visits by the sponsor to ensure protocol adherence and data integrity.
 - **Audit Trail:** Secure Electronic Data Capture (EDC) system featuring eCRF locking and full audit trails for data clarification forms.
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11. Study Identifiers & Governance

- **Primary ID:** CKJX839D12304
 - **Registry Number:** NCT05763875
 - **Posting ID:** 576102687985321766
 - **Sponsor/Lead Organization:** Novartis Pharmaceuticals
 - **Study Title:** Efficacy and Safety of Inclisiran as Monotherapy in Patients With Primary Hypercholesterolemia Not Receiving Lipid-lowering Therapy.
 - **Official Regulatory Title:** A Double-blind, Randomized, Placebo- and Active-Comparator Controlled Study to Evaluate the Efficacy of Inclisiran as Monotherapy in Patients With Primary Hypercholesterolemia Not Receiving Lipid-Lowering Therapy (VictORION-Mono)
 - **Trial Status:** Completed
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12. Clinical Framework (Hypercholesterolemia Specific)

- **Primary Condition:** Primary Hypercholesterolemia
 - **Disease Definition:** A condition with abnormally high levels of CHOLESTEROL in the blood. It is defined as a cholesterol value exceeding the 95th percentile for the population.
 - **Semantic Type:** Disease or Syndrome
 - **Preferred UMLS Name:** Drug Therapy for Hypercholesterolemia
 - **Ontology Codes:** MeSH Code: D006937 | SNOMED ID: 13644009 | HPO Code: HP:0003124
 - **Diagnosis Threshold:** Fasting LDL-C ≥ 100 mg/dL to < 190 mg/dL
 - **Medical Methodology:** siRNA-directed PCSK9 inhibition monotherapy
 - **Optimization Target:** Statistically significant percent reduction in LDL-C at Day 150 without concurrent statin background
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13. Study Procedures & Methodology

- **Management Pathway:** Subcutaneous monotherapy administration aligned with baseline cardiovascular risk assessment.
 - **Education Protocol (HCP):** Injection administration guidelines for subcutaneous siRNA dosing.
 - **Education Protocol (Patient):** General cardiovascular risk counseling, including foundational diet and exercise parameters.
 - **Quality Audit Mechanism:** Central lab evaluation to standardize LDL-C and PCSK9 assay reporting across all global sites.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Approximately 190 Days (150 \$\\pm\$ 5 day treatment phase + 30+5 day safety follow-up / End of Study visit)
 - **Onsite Visit Cadence:** Day -14 (Screening), Day 1, Day 90, Day 150, Day 180 (Approximate)
 - **Remote Monitoring Frequency:** Continuous AE tracking between the scheduled 3-month dosing intervals.
 - **Checklist Review Interval:** Full safety and concomitant medication review executed at every clinical visit.
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15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	____	____	[DD-MMM-YYYY]				
Clinical Operations Lead	__	__	[DD-MMM-YYYY]		Lead		
Statistician	__	____	[DD-MMM-YYYY]				